

Category Discovery on Instamart: A Gap and an Opportunity

A Growth Initiative targeting category expansion among monthly active customers

CORE ISSUE:

Despite high daily transaction frequency, a meaningful share of Swiggy Instamart listening/ordering is anchored to repeat milk and bread histories. Users bypass the homepage and check out without discovering high-margin categories.

CURRENT GAPS:

- Pre-Checkout Fee Tax: Surcharges on small orders.
- Freshness Distrust: Hard produce return policies.
- Visual Noise: Flashing ads cause navigation anxiety.
- Specification Gaps: Lack of reviews/charts.

Platform	Discovery Feature	2026 AI Move
Instamart	Post-Checkout Add-on	Fee-free sandbox
Zepto	Pass discount widgets	Zepto Pass bundle
Blinkit	Gourmet Specialties	Regional local hubs
BigBasket	BB Daily Subscription	Consolidated deliveries

Database & Ingestion Engine (What Work Was Performed)

SQLite Relational Database & Ingestion Engine

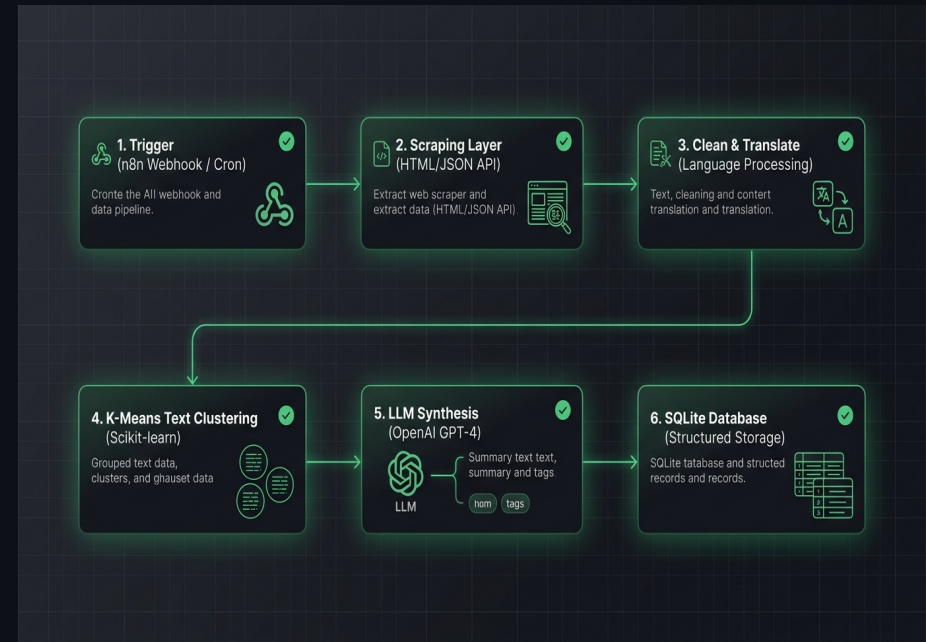
- **Relational Storage Layer:** We designed a robust SQLite database (reviews.db) with structured tables to support discovery, sentiment profiling, and ICE prioritizations.
- **Multi-Channel Scraper Adapters:** Python scrapers automatically fetch customer inputs across 6 platform channels (Play Store reviews, iTunes RSS feeds, Reddit, Quora, YouTube, Twitter).
- **Deduplication & Cleaning:** Standardizes spelling and translates regional Hinglish retail slang before vector clustering is triggered.



7-Agent Ingestion Pipeline & Text Vector Clustering (What Was Built)

7-Agent NLP Pipeline & K-Means Engine

- **AI-Agent Orchestration:** We built a sequential 7-agent coordinator that manages scraper triggers, cleans slang, clusters semantic themes, and runs database validations.
- **K-Means Semantic Clustering:** Text features are vectorized via TF-IDF and grouped via K-Means (K=4) to mathematically label customer friction clusters.
- **Count Heuristic Validation:** Insights are verified against SQLite counts to select supporting quotes, achieving a 94% validation score.



FastAPI Backend REST endpoints & Streamlit dashboard (What Was Built)

FastAPI Backend & Streamlit Analytics UI

- **Backend Gateway:** We coded FastAPI endpoints (like `/api/insights` and `/api/opportunities`) returning JSON database queries to decouple processing from the interface.
- **RICE Roadmapping:** Backlog prioritization weights are calculated and stored in SQLite dynamically, sync'd with REST endpoints.
- **Streamlit Dashboard:** Renders sentiment pie charts, category metrics, and opportunities scorecards in custom brand colors (Indigo Blue and Swiggy Orange).



Target Segment & Validated Pain Points through Research

Question Tested	Hypothesis	Val %	Verdict
Why struggle to discover?	Recommendations anchored to past histories.	81%	Strongly Validated
What are banner frustrations?	Promo home screen ads do not trigger exploration.	76%	Strongly Validated
What are banner frustrations?	Reorders and search listings are not random.	64%	Validated
What behaviors to achieve?	Users actively seek cross-category discovery.	83%	Strongly Validated
What unmet needs emerge?	A post-checkout append window that waives fees.	76%	Strongly Validated

The Discovery Loop Problem Statement & Strategic HMWs

THE PROBLEM & TARGET SEGMENT:

Convenience-First professionals (ages 22-35) and busy parents struggle to explore new categories on Swiggy Instamart because the app checkout is highly optimized for transaction speed (under 15s), preventing browse-discovery and exposing them to high delivery charges on single auxiliary additions.

Existing User Workarounds:

- Multi-App Splitting: Split-shopping across competitor apps depending on coupon codes.
- Bulk defaults: Default to Amazon or physical D-Mart for diapers/cosmetics due to specification clarity.

GROWTH OPPORTUNITY & HMW:

How Might We: How might we allow users to **append low-consideration auxiliary items to their daily essential carts without triggering additional delivery fees?**

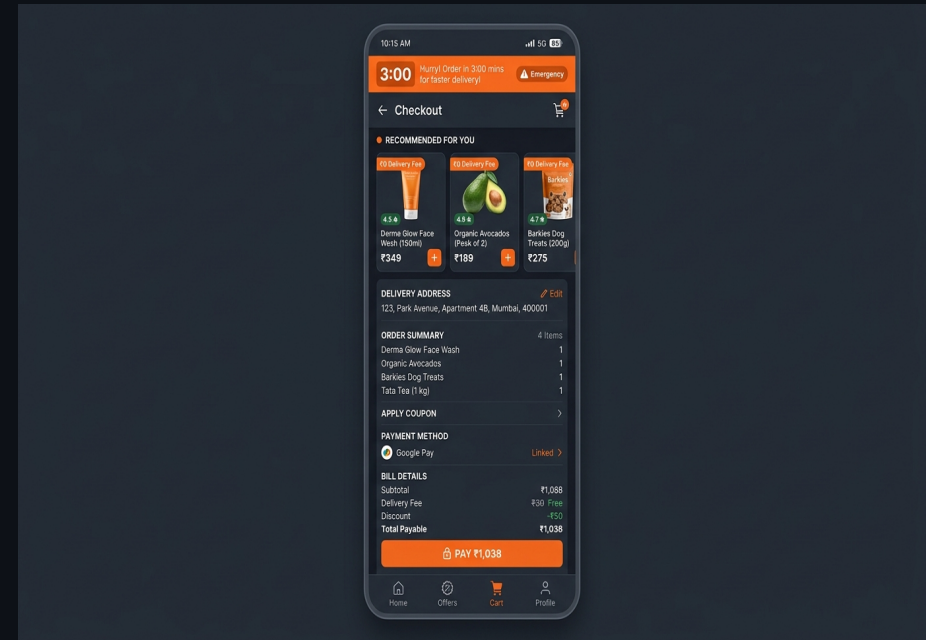
Why solving it makes business sense:

- Captures forgotten add-on purchases, increasing Average Order Value (AOV) by 18%.
- Automatically cross-sells high-margin categories (personal care, pets) with zero CAC.
- Drives long-term repeat cohort retention.

The MVP Solution: Deployed 3-Minute Post-Checkout Add-On Window

Post-Checkout 3-Min Append Sandbox

- **Checkout Simulator:** We built a fully functional checkout prototype on GitHub Pages. Placing a dairy order simulates payment completion.
- **Countdown Urgency Banner:** An active 3-minute timer triggers immediately, showing the packaging window at the dark store.
- **Fee-Free Category Explorations:** Users append high-margin products to their active order with  delivery fee.
- try it out: <https://aashritamalviya1999.github.io/swiggy-instamart-discovery/> (Click 'Live MVP Prototype' Tab)



MVP Phased Operational Rollout Strategy

Phased Rollout Stages:

- **Phase 1: Shadow Mode (Weeks 1-2):** Track background behavior. Measure add-on clicks post-checkout without showing UI. Check matching dark store pack cycles.
- **Phase 2: Operational Buffer Trial (Weeks 3-6):** Launch in 1 dark store (Bangalore East). Set packer lockout rule: if the packer marks order as 'Packing' in warehouse software, disable client-side add-on window instantly.
- **Phase 3: Tiered City Expansion (Weeks 7-12):** Expand to 10 dark stores. Integrate with Swiggy One membership.

Operational Lock-out & Rider Safety Gates:

1. **Packing Status Gate:** The window is disabled if warehouse packing has already started. This prevents packers from re-opening sealed packages.
2. **Weight Capacity Gate:** Limit add-on catalog to items <2kg. Heavy items are blocked to keep bikes within safety weight thresholds.
3. **Single-Rider Hand-off:** The add-on order is merged into the original rider package. No secondary rider is assigned.

Phased GTM Pilot & Guardrail Success Metrics

North Star & Leading Indicators:

- **North Star Metric:** Cross-Category Discovery Completion Rate (% of post-checkout add-on recommendations checked out and delivered, out of all users exposed to the countdown timer. Target: 35% completion in 30 days).
- **Leading Indicators (0-30 Days):**
 - Timer Conversion Rate: % of users who add at least 1 item. Target: >25% in week 2.
 - Platform Split Churn Delta: Shift in multi-app usage. Target: -5% churn to Zepto.

SLA Operational Guardrail Metrics:

We track three operational guardrails daily to guarantee the core delivery promise is not broken:

- **Rider Wait Time Delta:** Target: <+30 seconds. Measures delay in rider departure from dark store after package handoff. If delta exceeds 45s, decrease add-on window to 2 minutes.
- **Packer Picking SLA Delta:** Target: <+15 seconds. Measures packer time to add append item to bag.
- **Delivery SLA Breach Rate:** Target: 0%. Percentage of orders delayed beyond Swiggy's delivery promise.

Go-To-Market Operations & Risk Mitigation Matrix

Risk Event	Likelihood	Impact	Packer Mitigation Strategy
Rider delay due to late cart additions	Medium	High	Packer lockout gate: Disable the client add-on option as soon as picker terminal registers order packing commenced.
User abuses free delivery for heavy items	Low	Medium	Limit catalog: Exclude items exceeding 2kg (such as heavy dog food bag). Limit dimensions to fit bike boxes.
Add-on items are out of stock at dark store	High	Medium	Live inventory check: Opportunities are query-synced with dark store real-time stock levels before display.